

PRAKTIKUM SISTEM OPERASI

MODUL 5

EKSPLORASI PROSES XINU



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL

1. **[10 Poin]** Jawablah pertanyaan berikut ini:

a. Berapa banyaknya maksimum proses yang ada pada Xinu?

Jawaban: Maksimum jumlah proses pada Xinu adalah 8 proses, yang didefinisikan dengan konstanta NPROC = 8

b. Berapa maksimal panjang nama suatu proses pada Xinu?

Jawaban: 16 karakter define PNMLEN 16 termasuk karakter akhir (\0)

c. Berapa nilai prioritas awal pada saat proses dibuat?

Jawaban: 20 (Fungsi create() secara default menetapkan prioritas PR_DEFAULT = 20)

d. Ada berapa total state pada Xinu? Sebutkan!

Jawaban: 7 state

Yaitu:

1. PR_FREE: proses kosong / tidak digunakan
2. PR_CURR: sedang berjalan (running)
3. PR_READY: siap dijalankan
4. PR_RECV: menunggu pesan
5. PR_SLEEP: sedang tidur
6. PR_SUSP: ditangguhkan
7. PR_WAIT: menunggu semaphore

2. **[20 Poin]** Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari ps tersimpan pada file xsh_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!

Jawaban:

```
Activities Terminal Sat 11 Apr, 08:04
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ find / -name "xsh_.c" 2>/dev/null
xinu@xinu-develop-end:~$ ls
prev-version README xinu
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ ls
banner.h config DESCRIPTION include main.c shell
compile COPYRIGHT device lib net system
xinu@xinu-develop-end:~/xinu$ find . -name "xsh_ps.c"
./shell/xsh_ps.c
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_ps.c
1  /* xsh_ps.c - xsh_ps */
2
3  #include <xinu.h>
4  #include <stdio.h>
5  #include <string.h>
6
7  /*-----
8   * xsh_ps - shell command to print the process table
9   *-----
10  */
11  shellcmd xsh_ps(int nargs, char *args[])
12  {
13      struct procent *prptr;      /* Pointer to process */
14      int32 i;                    /* Index into proctabl */
15      char *pstate[] = {         /* Names for process states */
16          "free ", "curr ", "ready", "recv ", "sleep", "susp ",
17          "wait ", "rtime"};
18
```

```
Activities Terminal Sat 11 Apr, 08:04
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
19      /* For argument '--help', emit help about the 'ps' command */
20
21      if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
22          printf("Use: %s\n", args[0]);
23          printf("Description:\n");
24          printf("\tDisplays information about running processes\n");
25          printf("Options:\n");
26          printf("\t--help\t display this help and exit\n");
27          return 0;
28      }
29
30      /* Check for valid number of arguments */
31
32      if (nargs > 1) {
33          fprintf(stderr, "%s: too many arguments\n", args[0]);
34          fprintf(stderr, "Try '%s --help' for more information\n",
35                  args[0]);
36          return 1;
37      }
38
39      /* Print header for items from the process table */
40
41      printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
42             "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
43             "Stack Ptr", "Stack Size");
44
45      printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
46             "----", "-----", "-----", "----", "-----", "-----",

```

```
Activities Terminal Sat 11 Apr, 08:05
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
36         return 1;
37     }
38
39     /* Print header for items from the process table */
40
41     printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
42           "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
43           "Stack Ptr", "Stack Size");
44
45     printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
46           "----", "-----", "-----", "-----", "-----", "-----",
47           "-----", "-----", "-----");
48
49     /* Output information for each process */
50
51     for (i = 0; i < NPROC; i++) {
52         prptr = &proctab[i];
53         if (prptr->prstate == PR_FREE) { /* Skip unused slots */
54             continue;
55         }
56         printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
57               i, prptr->prname, prptr->prstate, prptr->prprio, prptr->prparent, prptr->prstkbase,
58               prptr->prstkptr, prptr->prstklen);
59     }
60     return 0;
61 }
62 }
xinu@xinu-develop-end:~/xinu$
```

```
Activities Terminal Sat 11 Apr, 08:31
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
GNU nano 2.2.6 File: ./shell/xsh ps.c Modified

}

/* Print header for items from the process table */

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "-----", "-----", "-----",
       "-----", "-----", "-----");

/* Output information for each process */

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
          i, prptr->prname, prptr->prstate, prptr->prprio, prptr->prparent, prptr->prstkbase,
          prptr->prstkptr, prptr->prstklen);
}

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

```
Activities Terminal Sat 11 Apr, 08:35
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
GNU nano 2.2.6 File: ./shell/xsh_ps.c Modified

}

/* Print header for items from the process table */
/* Menampilkan header/judul kolom untuk tabel proses*/

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "---", "-----", "-----", "-----", "-----", "-----",
       "-----", "-----");

/* Output information for each process */
/*Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5*/

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
           i, prptr->prname, pstate[(int)prptr->prstate],
           prptr->prprio, prptr->prparent, prptr->prstkbbase, prptr->prstkptr, prptr->prstklen);
}
return 0;
}

xinu@xinu-develop-end:~/xinu$
```

3. [35 Poin] Ubahlah perintah ps (source code: xsh_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses seperti gambar di bawah ini:

```
xsh $ ps
```

Pid	Name	State	Prio	Ppid	Stack Base	Stack Ptr	Stack Size	Msg	Content
0	prnull	ready	0	0	0x005FDFFC	0x00FFFF04	8192	1	4
1	rdspc	susp	200	0	0x005FBFFC	0x005FBFC8	16384	0	0
2	ipout	wait	500	0	0x005F7FFC	0x005F7F40	8192	0	0
3	netin	wait	500	0	0x005F5FFC	0x005F5EE0	8192	0	0
5	Main process	recv	20	4	0x005E3FFC	0x005E3F64	65536	0	0
6	shell	recv	50	5	0x005D3FFC	0x005D3C7C	8192	0	0
7	ps	curr	20	6	0x005F3FFC	0x005F3DC8	8192	0	0

Kolom Msg adalah banyaknya pesan yang ada dalam proses.

Kolom Content adalah isi dari pesan tersebut.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_ps.c

```

}

/* Print header for items from the process table */
/* Menampilkan header/judul kolom untuk tabel proses*/

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %8s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size", "Msg", "Content");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "-----", "-----", "-----",
       "-----", "-----", "-----", "-----");

/* Output information for each process */
/*Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5*/

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %8d\n",
           i, prptr->prname, prptr->prstate, prptr->prprio, prptr->ppid,
           prptr->prstackbase, prptr->prstackptr, prptr->prstacksize, prptr->prmsg, prptr->prcontent);
}

```

```

xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
GNU nano 2.2.6 File: ./shell/xsh_ps.c Modified

/* Output information for each process */
/*Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5*/

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %8d\n",
        i, prptr->prname, prptr->prstate, prptr->prprio, prptr->prparent, prptr->prstkbase,
        prptr->prstkptr, prptr->prstklen, prptr->prhasmsg, prptr->prmsg);
}
return 0;
}

```

- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$ps. Pastikan hasilnya sesuai dengan contoh output pada gambar yang diberikan.

```

xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

-----MUHAMMAD IKHSAN AL HAKIM-----
XINU
-----2311104064-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192 1      3
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192 0      0
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192 0      0
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34 65536 0      0
5 shell           recv   50   4 0x005F7FFC 0x005F79AC 8192 0      0
6 ps              curr   20   5 0x005F5FFC 0x005F5FC4 8192 0      0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0

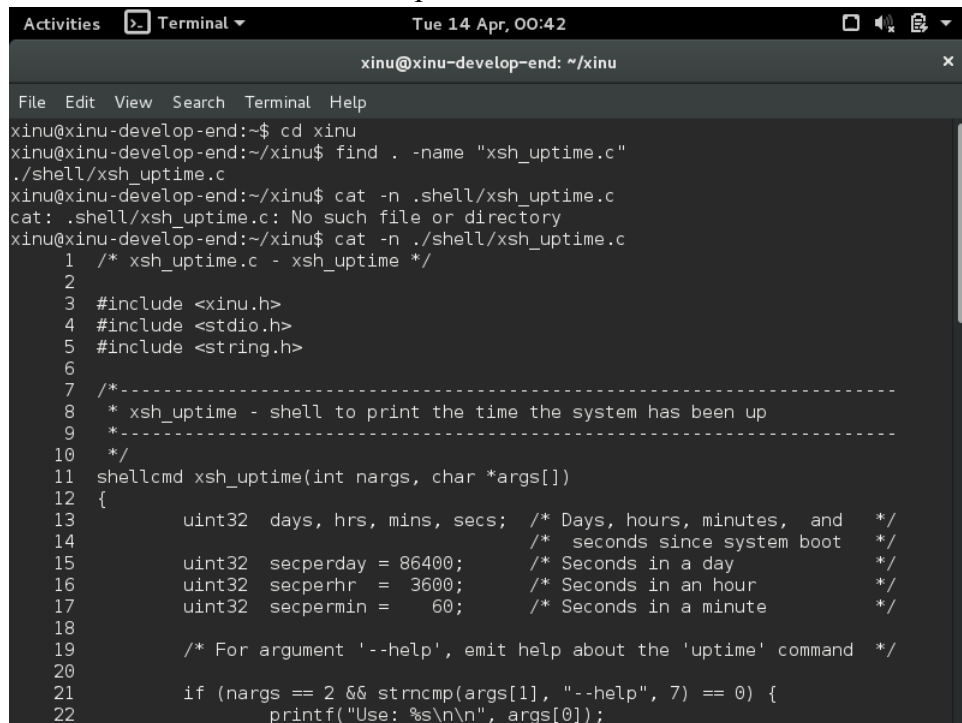
```

- Screenshot source code dan output akhir hasil modifikasi

4. **[35 Point]** Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_uptime.c
- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$uptime. Pastikan hasilnya sesuai dengan contoh output yang diinginkan
- Screenshot source kode dan output akhir hasil modifikasi



```
Activities Terminal Tue 14 Apr, 00:42
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ find . -name "xsh_uptime.c"
./shell/xsh_uptime.c
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_uptime.c
cat: ./shell/xsh_uptime.c: No such file or directory
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_uptime.c
 1  /* xsh_uptime.c - xsh_uptime */
 2
 3  #include <xinu.h>
 4  #include <stdio.h>
 5  #include <string.h>
 6
 7  /*-----
 8   * xsh_uptime - shell to print the time the system has been up
 9   *-----
10  */
11  shellcmd xsh_uptime(int nargs, char *args[])
12  {
13      uint32  days, hrs, mins, secs; /* Days, hours, minutes, and */
14                                     /* seconds since system boot */
15      uint32  secperday = 86400;    /* Seconds in a day */
16      uint32  secperhr  = 3600;     /* Seconds in an hour */
17      uint32  secpermin  = 60;      /* Seconds in a minute */
18
19      /* For argument '--help', emit help about the 'uptime' command */
20
21      if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
22          printf("Use: %s\n\n", args[0]);
```



```
Activities Terminal Tue 14 Apr, 00:48
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
GNU nano 2.2.6 File: ./shell/xsh_uptime.c Modified

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
    uint32 days, hrs, mins, secs; /* Days, hours, minutes, and */
                                /* seconds since system boot */
    uint32 secpday = 86400; /* Seconds in a day */
    uint32 secpshr = 3600; /* Seconds in an hour */
    uint32 secpmin = 60; /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
    }
}
```

```
Activities Terminal Tue 14 Apr, 00:47
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
GNU nano 2.2.6 File: ./shell/xsh_uptime.c Modified

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */

/* Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5*/
shellcmd xsh_uptime(int nargs, char *args[])
{
    uint32 days, hrs, mins, secs; /* Days, hours, minutes, and */
                                /* seconds since system boot */
    uint32 secpday = 86400; /* Seconds in a day */
    uint32 secpshr = 3600; /* Seconds in an hour */
    uint32 secpmin = 60; /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
    }
```

```
Activities Terminal Tue 14 Apr, 00:51
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
74         return 0;
75     }
xinu@xinu-develop-end:~/xinu$ nano ./shell/xsh_uptime.c
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...
xinu@xinu-develop-end:~/xinu/compile$ make
making the config program
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
flex config.l
bison -y          config.y
gcc -o config y.tab.c -lfl
./config Configuration conf.c conf.h
cp -p conf.h ../include
cp -p conf.c ../system
make[1]: Leaving directory '/home/xinu/xinu/config'
making ../system/conf.c
make[1]: Entering directory '/home/xinu/xinu/compile'
forcing a rebuild of conf.h and conf.c
make[2]: Entering directory '/home/xinu/xinu/compile'
creating new version
make[2]: Leaving directory '/home/xinu/xinu/compile'
```

```
Activities Terminal Tue 14 Apr, 00:51
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
[0x00100000 to 0x005FBFFF]
115657 bytes of Xinu code.
[0x00100000 to 0x0011C3C8]
148488 bytes of data.
[0x0011FEC0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)

-----MUHAMMAD IKHSAN AL HAKIM-----

XINU

-----2311104064-----

Welcome to Xinu!

xsh $ uptime
Xinu has been up 1 second(s)
xsh $
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

```
Activities Terminal Tue 14 Apr, 00:56 xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_uptime.c
1  /* xsh_uptime.c - xsh_uptime */
2
3  #include <xinu.h>
4  #include <stdio.h>
5  #include <string.h>
6
7  /*-----
8   * xsh_uptime - shell to print the time the system has been up
9   *-----
10 */
11 /* Aulia Jasifa Br Ginting - 2311104060 - Jurnal 5 */
12 shellcmd xsh_uptime(int nargs, char *args[])
13 {
14     uint32  days, hrs, mins, secs; /* Days, hours, minutes, and */
15                                     /* seconds since system boot */
16     uint32  secperday = 86400;      /* Seconds in a day */
17     uint32  secperhr  = 3600;       /* Seconds in an hour */
18     uint32  secpermin = 60;         /* Seconds in a minute */
19
20     /* For argument '--help', emit help about the 'uptime' command */
21
22     if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
23         printf("Use: %s\n\n", args[0]);
24         printf("Description:\n");
25         printf("\tDisplays time since the system booted\n");
26         printf("Options:\n");
27         printf("\t--help\t display this help and exit\n");
```